

Doc 8: Eigenvalues and Eigenvectors

Module 1: Linear algebra | Part 1 of 3: Learn (theory)

1. The idea: directions that survive

Most vectors change direction when multiplied by a matrix. An **eigenvector** v is special: $A v$ lands on the same line as v . Its only change is a stretch by a number λ (lambda), the **eigenvalue**:

$$A v = \lambda v \text{ (} v \text{ is not the zero vector)}$$

$\lambda > 1$ stretches, $0 < \lambda < 1$ shrinks, $\lambda < 0$ flips the direction, and $\lambda = 0$ squashes v to zero. Any nonzero multiple of an eigenvector is also an eigenvector.

$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$	$A v$	Verdict
$v = (1, 1)$	$(3, 3) = 3 \times (1, 1)$	eigenvector, $\lambda = 3$
$w = (1, -1)$	$(1, -1) = 1 \times (1, -1)$	eigenvector, $\lambda = 1$
$u = (1, 0)$	$(2, 1)$, not a multiple of $(1, 0)$	not an eigenvector

2. How to find them (2x2)

$A v = \lambda v$ means $(A - \lambda I) v = 0$. A nonzero solution exists only when **$\det(A - \lambda I) = 0$** . For a 2x2 matrix this becomes

$$\lambda^2 - (\text{trace}) \lambda + (\det) = 0 \text{ where trace} = a + d \text{ (the sum of the diagonal)}$$

- **Step 1.** Compute the trace and the determinant.
- **Step 2.** Solve the quadratic for λ : factor it, or use $\lambda = (t \pm \sqrt{t^2 - 4d}) / 2$. If $t^2 - 4d$ is negative there are no real eigenvalues.
- **Step 3.** For each λ , solve $(A - \lambda I) v = 0$. The two equations are multiples of each other, so use just one row.
- **Step 4.** Check: $A v$ should equal λv .

Sum of eigenvalues = trace

a quick check on your answers

Product of eigenvalues = determinant

so $\det = 0$ exactly when 0 is an eigenvalue

Triangular or diagonal eigenvalues are the diagonal entries

$\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$ has $\lambda = 3$ and 2

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Part 1 of 3: Learn (theory), continued

3. Diagonalization: the best basis

Suppose A has two independent eigenvectors. Put them in the columns of V and the eigenvalues on the diagonal of D . Then $A V = V D$, so

$$A = V D V^{-1} \text{ and } A^n = V D^n V^{-1}$$

Read it right to left, using Doc 7: V^{-1} converts a vector to eigen-coordinates, D stretches each coordinate on its own, and V converts back. In the eigenbasis, A is a plain stretch. That is why powers are easy: only the diagonal numbers get powered, so $D^n = \text{diag}(\lambda_1^n, \lambda_2^n)$.

4. Special cases worth knowing

Symmetric matrix ($A^T = A$)	real eigenvalues; eigenvectors are perpendicular	$[2 \ 1; 1 \ 2]: (1, 1) \cdot (1, -1) = 0$
Rotation	no real eigenvectors	every direction turns (page 3, right picture)
Repeated eigenvalue	may have only one eigenvector direction	$[2 \ 1; 0 \ 2]$ cannot be diagonalized
Zero eigenvalue	A is singular ($\det = 0$)	some direction is squashed to 0

5. Where machine learning uses eigenvalues

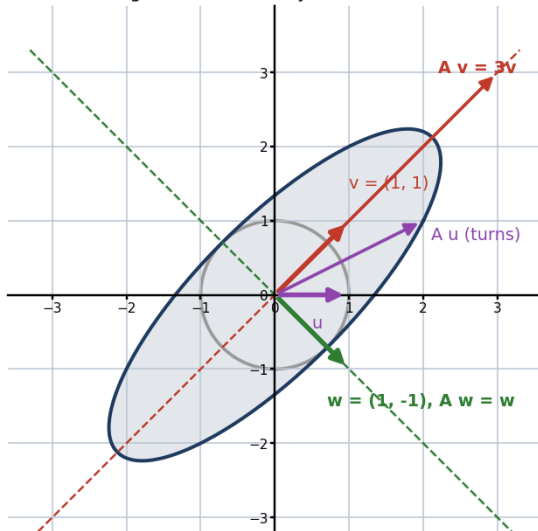
- **PCA** (Doc 9): the eigenvectors of the covariance matrix are the principal directions; the eigenvalues are the variances along them.
- **Repeated multiplication** (RNNs, Doc 27): $A^n x$ grows like λ^n . If $|\lambda| > 1$ values explode; if $|\lambda| < 1$ they vanish. This is the exploding and vanishing gradient problem.
- **Power iteration and PageRank**: multiplying by A again and again lines a vector up with the dominant eigenvector (Example 5).
- **Optimization** (Docs 12-13): the eigenvalues of the Hessian describe curvature; a big gap between the largest and smallest makes gradient descent slow.
- **SVD** (Doc 9): the singular values of A come from the eigenvalues of $A^T A$.

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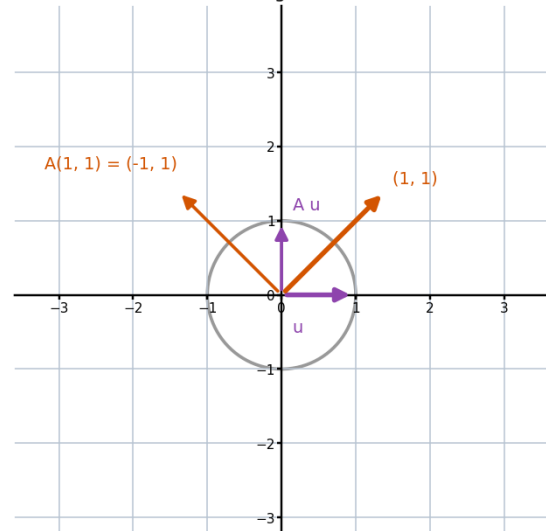
Part 2 of 3: Worked examples. Every cell filled in. Read, then cover the answers and redo them.

Example 1: Which vectors keep their direction?

$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$: circle becomes an ellipse.
Eigen-directions stay on their lines



Rotation by 90 degrees: every vector turns.
No real eigenvectors



		1	1	1
		1	-1	0
2	1	3	1	2
1	2	3	-1	1

$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ applied to three vectors (the columns of the top matrix): $(1, 1)$, $(1, -1)$, $(1, 0)$.

Results (columns of C): $(3, 3) = 3 \times (1, 1)$ and $(1, -1) = 1 \times (1, -1)$: both stay on their lines. $(2, 1)$ is not a multiple of $(1, 0)$: it turned.

Example 2: Finding eigenvalues and eigenvectors of $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$

Step 1. trace = $4 + 3 = 7$, det = $4 \times 3 - 1 \times 2 = 10$. $A V$ (V has the eigenvectors as columns)

$V D$ ($D = \text{diag}(5, 2)$)

Step 2. $\lambda^2 - 7\lambda + 10 = 0$, so $(\lambda - 5)(\lambda - 2) = 0$, giving $\lambda = 5$ and $\lambda = 2$.

Step 3. $\lambda = 5$: $A - 5I = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$. Row 1: $-x + y = 0$, so $v = (1, 1)$.

$\lambda = 2$: $A - 2I = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$. Row 1: $2x + y = 0$, so $v = (1, -2)$.

Step 4. $A(1, 1) = (5, 5) = 5(1, 1)$. $A(1, -2) = (2, -4) = 2(1, -2)$.

		1	1
		1	-2
4	1	5	2
2	3	5	-4

		5	0
		0	2
1	1	5	2
1	-2	5	-4

Both grids are equal: $A V = V D$. That single equation says every column of V is an eigenvector. Check: $5 + 2 = 7 = \text{trace}$ and $5 \times 2 = 10 = \text{det}$.

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Part 2 of 3: Worked examples, continued

Example 3: The triangular shortcut

$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$ is triangular, so its eigenvalues are the diagonal entries: $\lambda = 3$ and $\lambda = 2$. $\lambda = 3$: $A - 3I = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$, so $y = 0$ and $\mathbf{v} = (1, 0)$. $\lambda = 2$: $A - 2I = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, so $x + y = 0$ and $\mathbf{v} = (1, -1)$. Check: $A(1, -1) = (3 - 1, -2) = (2, -2) = 2(1, -1)$.

Example 4: Diagonalization and powers, $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

Eigenvalues 3 and 1 with eigenvectors $(1, 1)$ and $(1, -1)$ give $V = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $D = \text{diag}(3, 1)$, $V^{-1} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix}$.

AV	VD	$(VD^2)V\text{-inverse} = A^2$	$(VD^5)V\text{-inverse} = A^5$																																																																
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First two grids: $AV = VD$. Third: $D^2 = \text{diag}(9, 1)$, and $VD^2 = \begin{bmatrix} 9 & 1 \\ 9 & -1 \end{bmatrix}$ times V^{-1} gives $A^2 = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$. Fourth: $D^5 = \text{diag}(243, 1)$ gives $A^5 = \begin{bmatrix} 122 & 121 \\ 121 & 122 \end{bmatrix}$ without multiplying A five times.

Example 5: Power iteration: repeated multiplication finds the dominant eigenvector

Start with $\mathbf{x} = (1, 0)$ and multiply by $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ again and again.

n	$A^n \mathbf{x}$	second $\div$ first	first $\div$ previous first
0	(1, 0)	0	-
1	(2, 1)	0.5	2
2	(5, 4)	0.8	2.5
3	(14, 13)	0.929	2.8
4	(41, 40)	0.976	2.929

The ratio second $\div$ first heads to 1, so the direction approaches $(1, 1)$, the dominant eigenvector. The growth per step heads to 3, the largest eigenvalue. If that number were below 1 the vector would shrink to nothing.

Quick reference card

Definition	$A\mathbf{v} = \lambda\mathbf{v}$, $\mathbf{v} \neq 0$	eigenvector $\mathbf{v}$, eigenvalue λ
2×2 eigenvalues	$\lambda^2 - (\text{trace})\lambda + (\text{det}) = 0$	sum = trace, product = det
Eigenvectors	solve $(A - \lambda I)\mathbf{v} = 0$	use one row; check $A\mathbf{v} = \lambda\mathbf{v}$
Diagonalize	$A = VDV^{-1}$, $A^n = VD^nV^{-1}$	V = eigenvectors, D = eigenvalues

Doc 8: Eigenvalues and Eigenvectors

Part 3 of 3: Practice. Use a pencil; yellow cells are yours. Name: _____ Date: _____

A. Is it an eigenvector? $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

Compute A times each vector (the columns of the top matrix). Then decide: is $A v$ a multiple of v ? If so, what is λ ?

		1	1	1	2
		1	-1	0	2
1	2				
2	1				

Vector	Eigenvector?	λ
(1, 1)		
(1, -1)		
(1, 0)		
(2, 2)		

B. Eigenvalues from the trace and determinant

Solve $\lambda^2 - (\text{trace})\lambda + (\text{det}) = 0$ for each matrix. Check: the eigenvalues add to the trace and multiply to the determinant.

Matrix	trace	det	Eigenvalues
$\begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$			
$\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$			
$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$			
$\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$			

C. Full solve for $A = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$ (eigenvalues 6 and 1)

Write $A - \lambda I$, solve for v from one row, then check that $A v = \lambda v$ using the grid.

$A - 6I$

eigenvector v

Check: $A v$ (write v on top) $\lambda = 6$

5	2	
2	2	

$$A v = \underline{\hspace{2cm}} \times v$$

$A - 1I$

eigenvector v

Check: $A v$ (write v on top) $\lambda = 1$

5	2	
2	2	

$$A v = \underline{\hspace{2cm}} \times v$$

Doc 8: Eigenvalues and Eigenvectors

Practice, continued

D. The triangular shortcut

Matrix	Eigenvalues	Eigenvector for the smaller eigenvalue
[4 1; 0 3]		
[2 0; 5 3]		

E. Diagonalization: $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, $V = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $D = \text{diag}(3, -1)$

$V^{-1} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix}$. Check $A V = V D$, then use $A^n = V D^n V^{-1}$.

$A V$	$V D$	$A^2 = (V D^2) V^{-1}$	$A^5 = (V D^5) V^{-1}$																																																																
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Are $A V$ and $V D$ equal? _____. For A^5 the top-left matrix is $V D^5 = \begin{bmatrix} 243 & -1 \\ 243 & 1 \end{bmatrix}$ because $D^5 = \text{diag}(243, -1)$.

F. Power iteration: $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$, $x = (1, 0)$

n	$A^n x$ (multiply the previous vector by A)	second ÷ first (3 decimals)
1		
2		
3		
4		

The direction approaches (____, ____). Growth estimate: $136 \div 36 = \underline{\hspace{2cm}}$, heading toward the largest eigenvalue ____.

Doc 8: Eigenvalues and Eigenvectors

Practice, continued

G. Machine learning: vanishing and exploding

$A = \begin{bmatrix} 0.5 & 0 \\ 0 & 1.5 \end{bmatrix}$ is diagonal, so its eigenvalues are 0.5 and 1.5. Apply it repeatedly to $x = (1, 1)$.

n	$A^n x$
1	
2	
3	

Eigenvalues: _____ and _____. Which component shrinks toward 0? _____ Which grows without bound? _____

PCA link: a covariance matrix $C = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has eigenvalues 3 and 1 (Example 1). The data vary most along the direction (____, ____).

H. What does the eigenvalue do to the eigenvector?

Eigenvalue	What happens to v ?
$\lambda = 2$	
$\lambda = 1$	
$\lambda = 0.5$	
$\lambda = 0$	
$\lambda = -1$	
$\lambda = -3$	

I. Challenge: solve a new matrix from scratch, $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$

Step	Your answer
trace and det	
eigenvalues	
eigenvectors (for 5, then for 2)	

Now check $AV = VD$: write your eigenvectors as the columns of V , and your eigenvalues on the diagonal of D .

AV

3	2		
1	4		

VD

Are AV and VD equal? _____

Doc 8: Eigenvalues and Eigenvectors

Answer key (page 1 of 2): check your work only after finishing

A. Is it an eigenvector? $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

		1	1	1	2
		1	-1	0	2
1	2	3	-1	1	6
2	1	3	1	2	6

Vector	Eigenvector?	λ
(1, 1)	Yes	3
(1, -1)	Yes	-1
(1, 0)	No	none
(2, 2)	Yes	3

$A \cdot v$: $(3, 3) = 3(1, 1)$; $(-1, 1) = -1(1, -1)$; $(1, 2)$ is not a multiple of $(1, 0)$; $(6, 6) = 3(2, 2)$.

B. Eigenvalues from the trace and determinant

Matrix	trace	det	Eigenvalues
$\begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$	7	6	1 and 6
$\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$	3	-4	4 and -1
$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$	4	4	2 (twice)
$\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$	-3	2	-1 and -2

$\begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$: $\lambda^2 - 7\lambda + 6 = (\lambda - 1)(\lambda - 6)$. $\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$: $\lambda^2 - 3\lambda - 4 = (\lambda - 4)(\lambda + 1)$. $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$: $(\lambda - 2)^2$. $\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$: $\lambda^2 + 3\lambda + 2 = (\lambda + 1)(\lambda + 2)$.

C. Full solve for $A = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$

$$\lambda = 6: A - 6I$$

-1	2
2	-4

$$v = (2, 1)$$

2
1

$$A \cdot v$$

		2
		1
5	2	12
2	2	6

$$\lambda = 1: A - I$$

4	2
2	1

$$v = (1, -2)$$

1
-2

$$A \cdot v$$

		1
		-2
5	2	1
2	2	-2

$\lambda = 6$: $-x + 2y = 0$ gives $v = (2, 1)$, and $A \cdot v = (12, 6) = 6v$. $\lambda = 1$: $4x + 2y = 0$ gives $v = (1, -2)$, and $A \cdot v = (1, -2) = 1v$. Any multiple of v is also fine.

D. The triangular shortcut

Matrix	Eigenvalues	Eigenvector for the smaller eigenvalue
$\begin{bmatrix} 4 & 1 \\ 0 & 3 \end{bmatrix}$	4 and 3	(1, -1)
$\begin{bmatrix} 2 & 0 \\ 5 & 3 \end{bmatrix}$	2 and 3	(1, -5)

$\begin{bmatrix} 4 & 1 \\ 0 & 3 \end{bmatrix}$: $\lambda = 3$ gives $A - 3I = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, so $x + y = 0$. $\begin{bmatrix} 2 & 0 \\ 5 & 3 \end{bmatrix}$: $\lambda = 2$ gives $A - 2I = \begin{bmatrix} 0 & 0 \\ 5 & 1 \end{bmatrix}$, so $5x + y = 0$.

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Answer key (page 2 of 2)

E. Diagonalization

A V

		1	1
		1	-1
1	2	3	-1
2	1	3	1

V D

		3	0
		0	-1
1	1	3	-1
1	-1	3	1

A²

		0.5	0.5
		0.5	-0.5
9	1	5	4
9	-1	4	5

A⁵

		0.5	0.5
		0.5	-0.5
243	-1	121	122
243	1	122	121

$A V = V D = [3 \ -1; 3 \ 1]$: yes, equal. $A^2 = [5 \ 4; 4 \ 5]$ and $A^5 = [121 \ 122; 122 \ 121]$.

F. Power iteration

n	A ⁿ x	second ÷ first
1	(3, 1)	0.333
2	(10, 6)	0.6
3	(36, 28)	0.778
4	(136, 120)	0.882

Direction approaches (1, 1). Growth $136 \div 36 = 3.78$, heading toward the largest eigenvalue 4. ($A = [3 \ 1; 1 \ 3]$ has $\lambda = 4$ for (1, 1) and $\lambda = 2$ for (1, -1).)

G. Vanishing and exploding

n	A ⁿ x
1	(0.5, 1.5)
2	(0.25, 2.25)
3	(0.125, 3.375)

Eigenvalues 0.5 and 1.5. The first component shrinks toward 0 (0.5^n); the second grows without bound (1.5^n). PCA link: the data vary most along (1, 1), where the eigenvalue (variance) is 3.

H. What the eigenvalue does

Eigenvalue	What happens to v?
$\lambda = 2$	stretched to twice as long, same direction
$\lambda = 1$	unchanged
$\lambda = 0.5$	shrunk to half as long, same direction
$\lambda = 0$	squashed to the zero vector
$\lambda = -1$	flipped to the opposite direction, same length
$\lambda = -3$	flipped and stretched to 3 times as long

I. Challenge

Step	Answer	A V	V D
trace and det	7 and 10		
eigenvalues	5 and 2		
eigenvectors (for 5, then for 2)	(1, 1) and (2, -1)		

		1	2
		1	-1
3	2	5	4
1	4	5	-2

		5	0
		0	2
1	2	5	4
1	-1	5	-2

$\lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2)$. $\lambda = 5$: $A - 5I = [-2 \ 2; 1 \ -1]$, so $v = (1, 1)$. $\lambda = 2$: $A - 2I = [1 \ 2; 1 \ 2]$, so $x + 2y = 0$ and $v = (2, -1)$. $A V = V D = [5 \ 4; 5 \ -2]$.

Key takeaways: $A v = \lambda v$: eigenvectors keep their direction and are stretched by λ ; for 2×2 , solve $\lambda^2 - (\text{trace}) \lambda + (\text{det}) = 0$, then $(A - \lambda I) v = 0$; $A = V D V^{-1}$ makes powers easy; $|\lambda|$ above 1 explodes and below 1 vanishes; symmetric matrices have perpendicular eigenvectors.

Next: Doc 9 - SVD, low-rank approximation and PCA.